

Winning Back the Base

A Value-Based Customer Retention Strategy for Company A

Turning 100,000 customer records into a quantified, ROI-positive retention program — targeting not just who will churn, but who is most valuable to keep.

0.692

XGBoost ROC-AUC (CV-validated)

\$10.1M

net benefit / yr (per 1M subs)

1.6x

more revenue saved vs. churn-only

Executive summary

Company A loses customers faster than it can profitably replace them. We can fix that.

■ The problem

In a saturated market, ~95% of growth comes from switchers and re-acquiring a lost customer costs 5-25x more than keeping one. Churn is Company A's most expensive leak.

■ What the data says

Churn is predictable and concentrated. The strongest driver is aging handsets - devices past ~12 months churn at 54-58% vs ~42% when new - a lever Company A controls.

■ Our solution

A monthly value-based targeting engine: rank customers by Expected Value-at-Risk ($P(\text{churn}) \times \text{customer value}$), then make proactive, persona-specific retention offers.

What it delivers - per 1,000,000 subscribers

Best model	XGBoost · AUC 0.692
Targeting edge	captures 44% of revenue-at-risk in top 20%
Net benefit / yr	\$10.1M (1.68x ROI)
Customers saved / yr	~16,949
Our fee	\$840k/yr · 8.3% of value created

Sources: Bain & Company; Harvard Business Review; analysis of Company A dataset (n=100,000).

Market analysis: retention is the only profitable growth lever

\$344B

US wireless market (2025)

1.0%

Best-in-class monthly churn

5-25x

Re-acquire vs. retain cost

25%+

Profit lift from +5% retention

■ **The market is saturated**

Penetration exceeds 100%. Carriers grow almost entirely by stealing switchers - making every avoided cancellation worth far more than a new-customer ad.

■ **Acquisition is expensive**

Industry CAC runs into the hundreds of dollars per line; retention offers are a fraction of that, so saved customers are immediately accretive.

■ **Small churn moves, big profit**

Bain's classic finding: a 5-point retention improvement can lift profits 25-95%. Churn is the highest-leverage metric in the P&L.

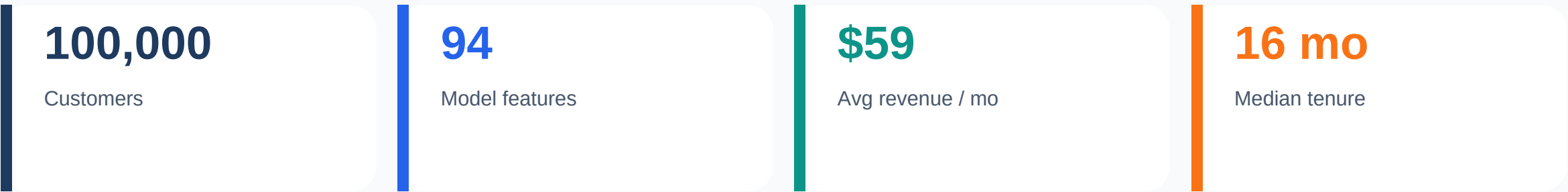
■ **Loyalty compounds**

Tenured customers spend more, cost less to serve, and refer others - so protecting the base protects future revenue too.

Sources: industry market sizing 2025; Verizon/T-Mobile investor reports; Bain & Company; Reichheld, HBR.

The client & the data

Company A is a US wireless carrier facing elevated churn. We received two linked tables covering 100,000 customers - behavioural usage and customer profile - joined on Customer_ID.



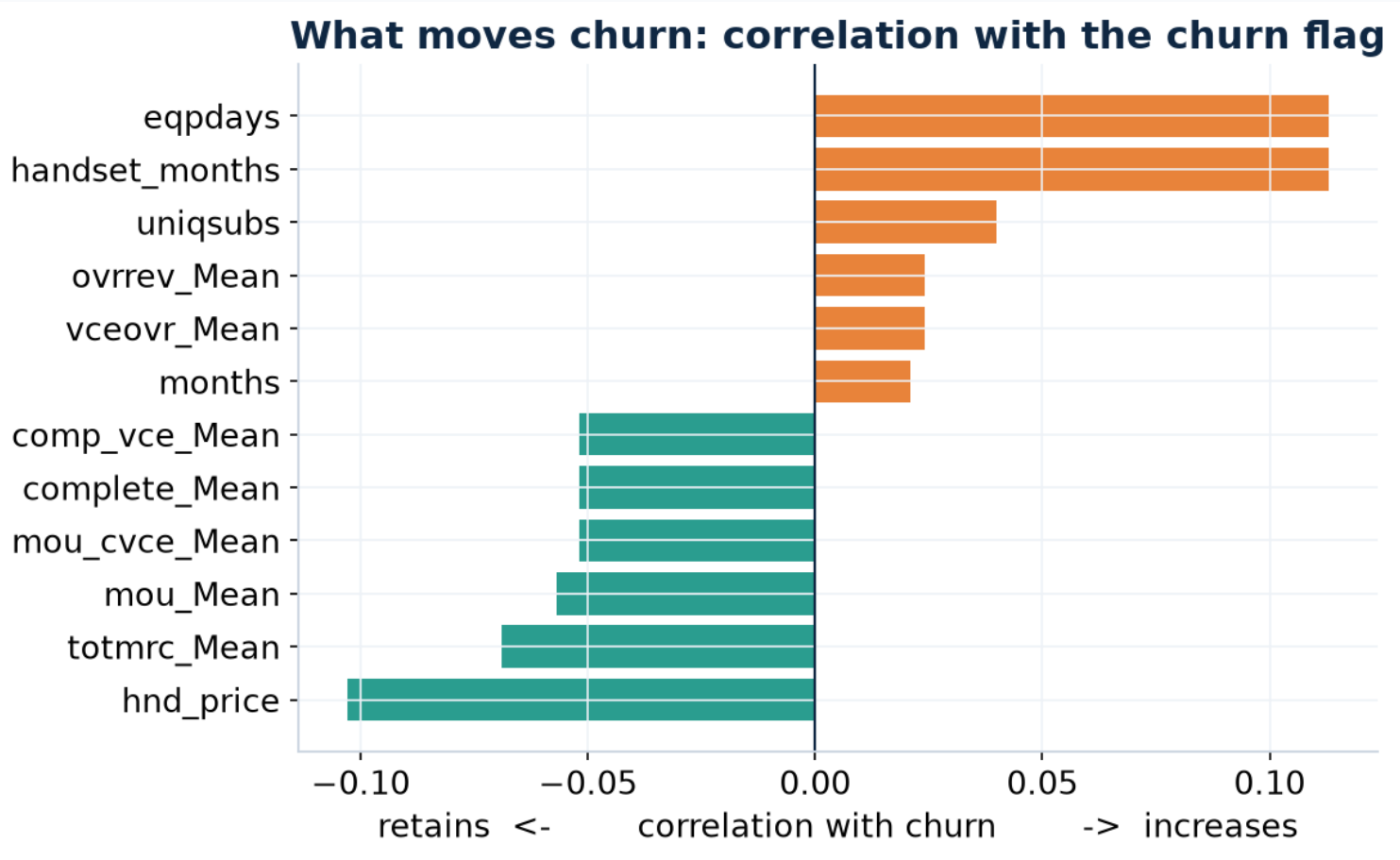
Behavioural (Record.csv)
Minutes of use, overage, revenue, call quality, recharges, and the churn label - the in-life signals of disengagement.

Profile (Client.csv)
Handset age & price, refurbished flag, web-capability, tenure - the structural drivers we can act on.

Data discipline: (1) the modelling sample is balanced ~50/50 by design (oversampled), so it is used for ranking - the real operating churn assumption (22%/yr) is applied separately in the business case; (2) we DROP 9 fields with 25-49% missingness and 4 protected attributes (ethnicity, marital status, income, credit class) - see slide 15 for the fairness test.

Source: Company A dataset (Client.csv + Record.csv), 100,000 customers.

What the data reveals: who leaves, and why



Equipment age is the #1 signal

Time on the same handset (eqpdays / handset_months) is the strongest positive correlate of churn (+0.11) - phones age out and customers shop around.

Investment buys loyalty

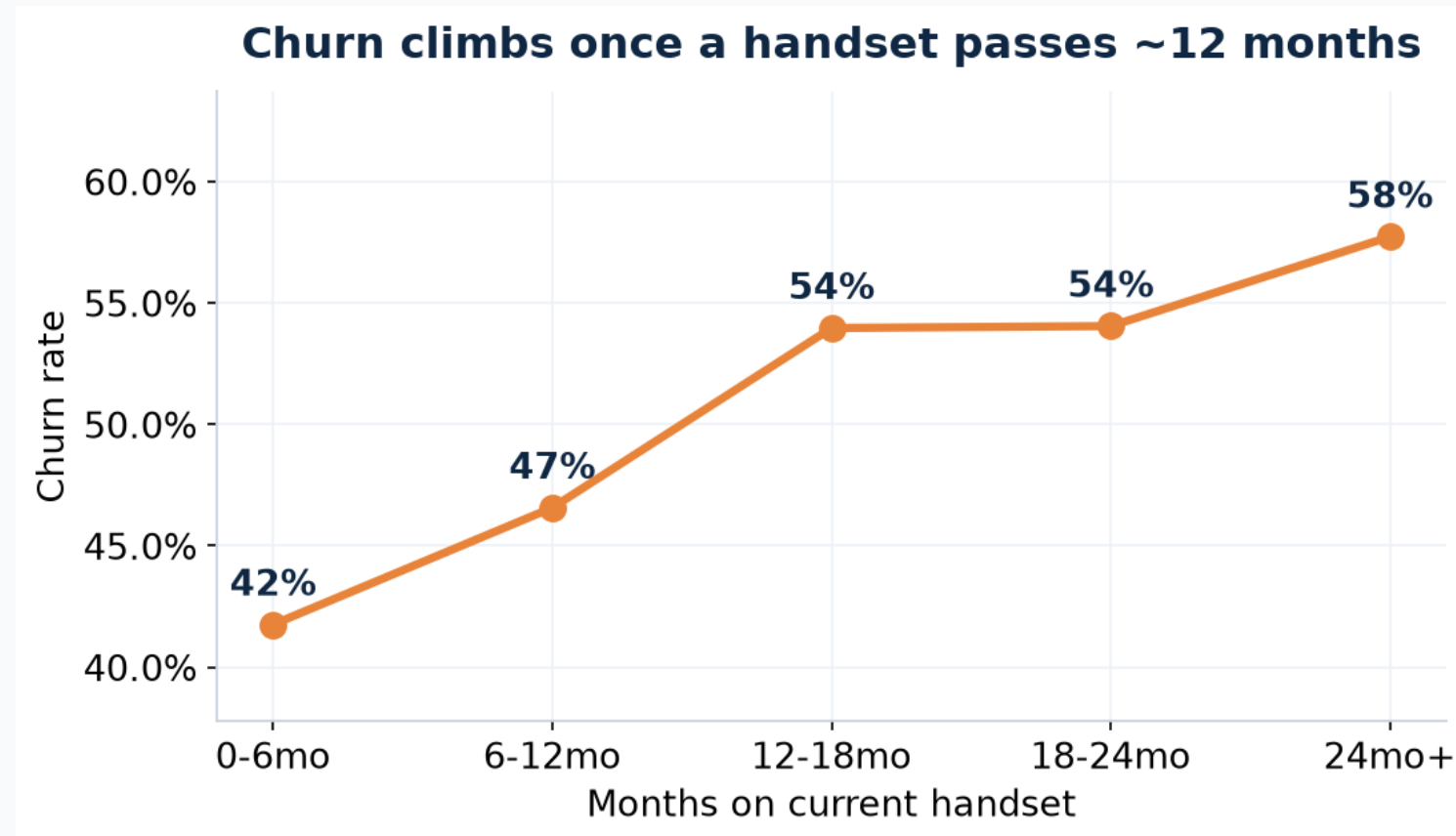
Higher handset price and higher recurring charge both correlate negatively with churn - customers with newer, pricier phones stay.

Engagement matters

More minutes of use and completed calls track with lower churn; a declining usage trend is an early-warning sign we engineered into the model.

Source: point-biserial correlations with churn flag, Company A dataset (n=100,000).

The headline insight: the handset-age cliff



- A clear climb after ~12 months**

Churn rises from ~47% (6-12 mo) to ~54% past 12 months and reaches ~58% beyond two years.

- This is an actionable lever**

Unlike age or income, handset age is something Company A can change - through proactive upgrade and trade-in offers timed to the cliff.

- Real months, not guesses**

Buckets are computed directly from days-on-handset in the data - no assumed labels - so the timing of the intervention is evidence-based.

Source: churn rate by months-on-current-handset, Company A dataset.

Framing the business problem as an ML task

Business question: 'Which customers are about to leave, how much revenue do they put at risk, and what is the most cost-effective way to keep them?'

■ ML formulation

Supervised binary classification: predict $P(\text{churn})$ per customer from behavioural + profile features, then calibrate to the real base rate.

■ But probability isn't enough

A \$149/mo customer and a \$42/mo customer are not equally worth saving. We rank by Expected Value-at-Risk = $P(\text{churn}) \times \text{customer value}$.

■ Primary metric: ROC-AUC

Ranking quality matters more than a single cutoff, because we act on the highest-risk, highest-value customers first.

■ Success = profit, not accuracy

Judged by incremental margin + avoided re-acquisition cost from a targeted campaign, validated by an A/B holdout.

Modelling approach: rigorous, reproducible, honest

01

Clean & engineer

Drop 9 high-missing + 4 protected fields; impute, encode; engineer 7 features (usage trend, overage share, handset months...). 94 model features.

02

Train 4 model families

Logistic Regression (baseline), Random Forest, XGBoost and LightGBM - from interpretable to high-performance.

03

Validate & calibrate

75/25 stratified split + 5-fold CV on the winner; probabilities calibrated and prior-shifted to the 22% operating base rate.

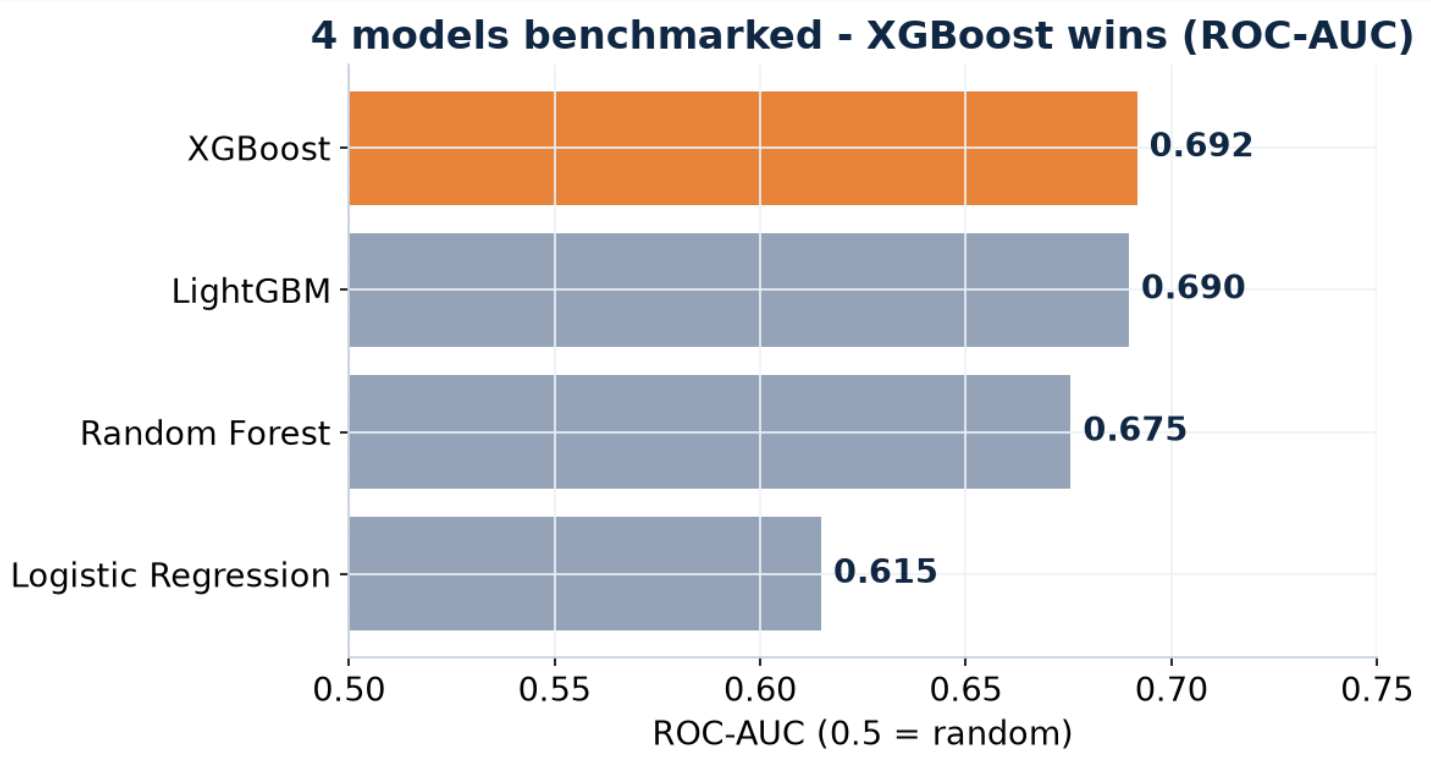
04

Rank by value

Convert to Expected Value-at-Risk; profit-curve sets the contact threshold; simulate vs random and churn-only baselines.

Cross-validation result: XGBoost ROC-AUC = 0.695 +/- 0.003 across 5 folds - confirming the model generalises and the score is stable, not luck.

Results: XGBoost wins - AUC 0.692

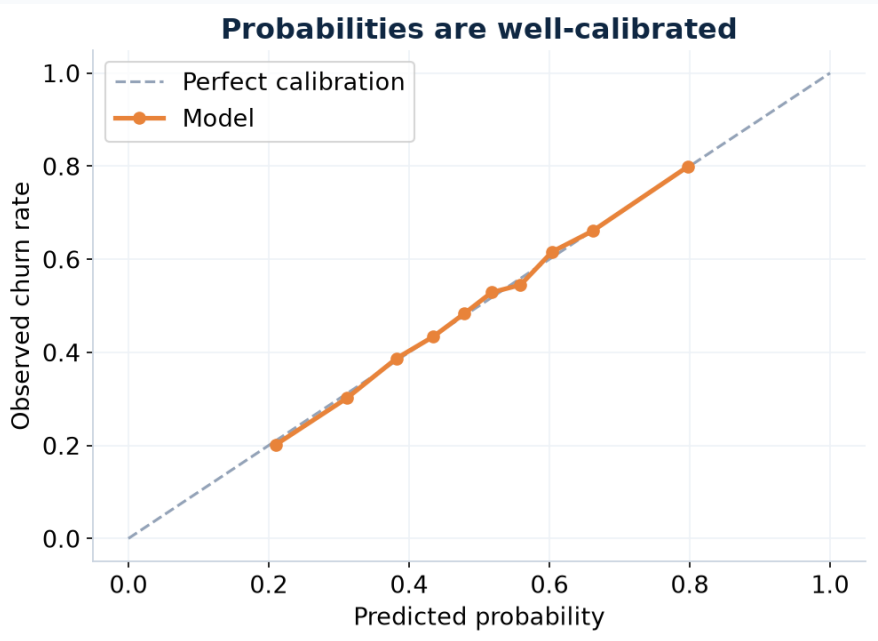
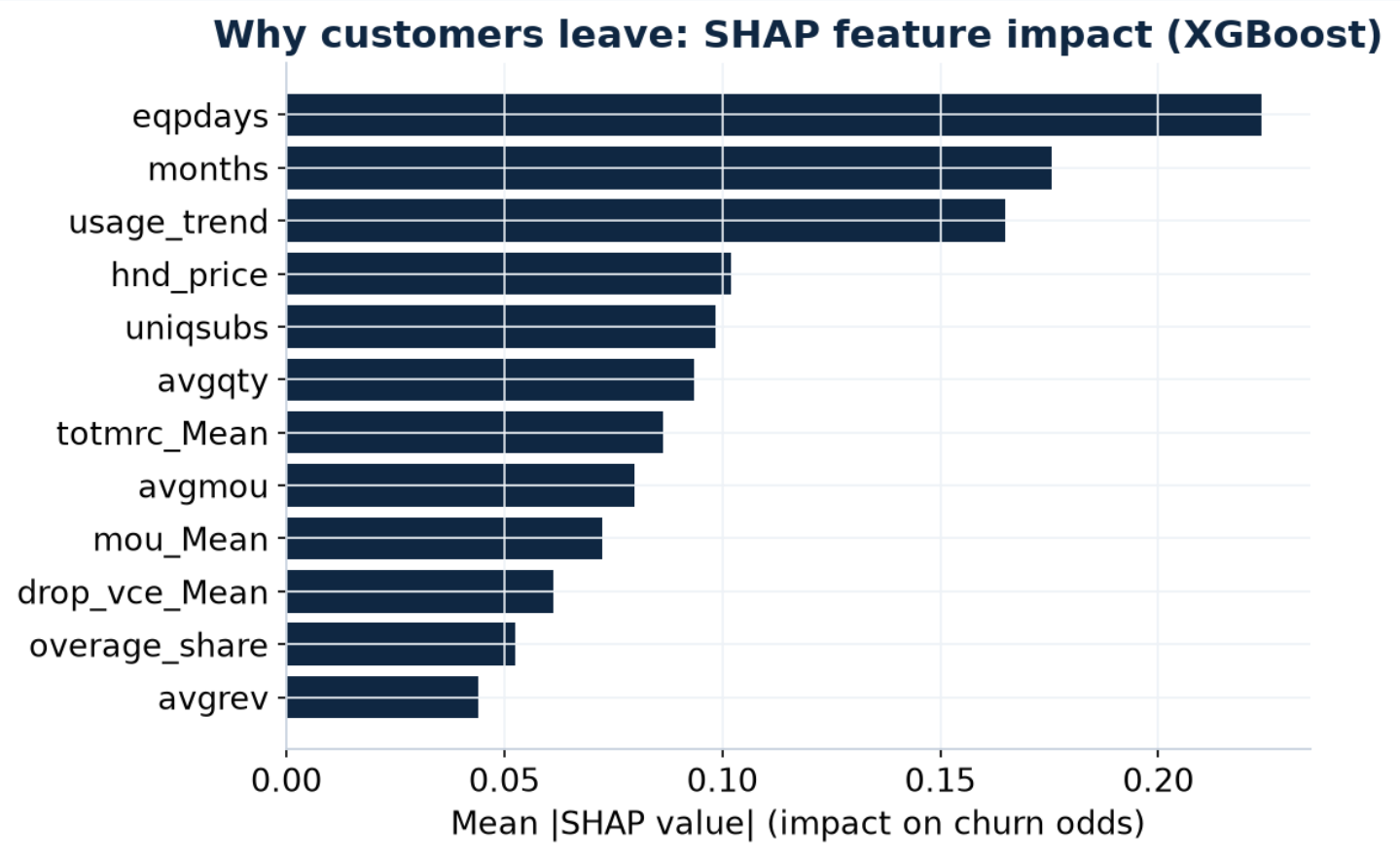


Model	AUC	Acc	Prec	Recall
Logistic Regression	0.615	0.586	0.585	0.563
Random Forest	0.675	0.619	0.612	0.630
LightGBM	0.690	0.633	0.630	0.632
XGBoost	0.692	0.634	0.631	0.631

How to read AUC: 0.5 = guessing, 1.0 = perfect. At 0.692 the model ranks a churner above a non-churner ~69% of the time - enough to target retention spend profitably. Both gradient-boosted models clearly beat the linear baseline.

Source: held-out test set (25% of 100,000), Company A dataset.

Why we trust the model: explainable & calibrated



Explainable (SHAP)

The top churn drivers are eqpdays, months and usage_trend - all behaviour Company A can act on. No protected attribute appears.

Calibrated

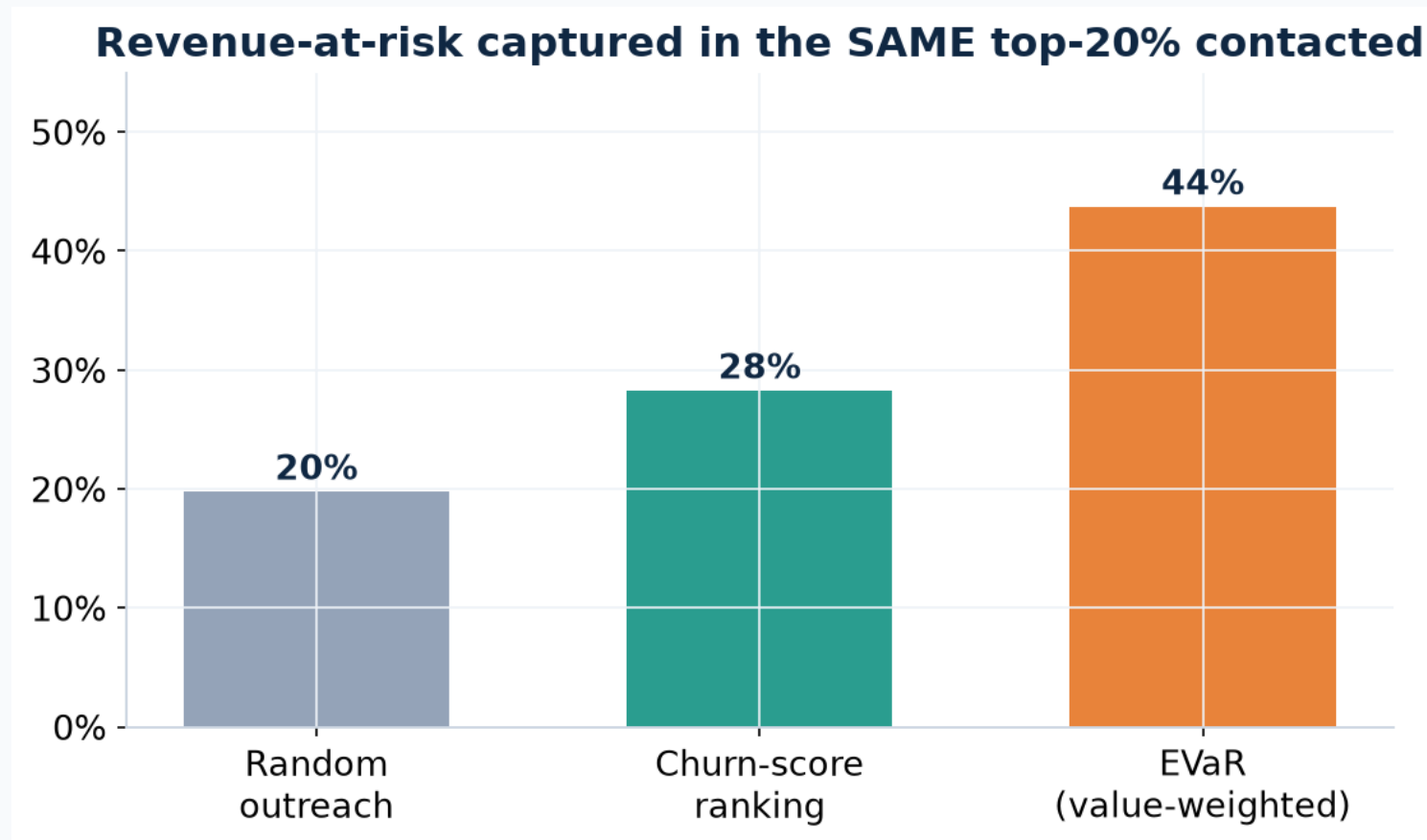
Predicted probabilities track observed churn along the 45deg line, so EVaR dollar figures are trustworthy - not just rankings.

Actionable

Because drivers are controllable, every score maps to a concrete offer.

Source: SHAP values on held-out set; reliability curve (10 bins) after prior-shift calibration.

Our differentiator: target value, not just probability



The insight

Ranking by churn probability alone wastes budget on low-value churners. Ranking by Expected Value-at-Risk protects the revenue that actually matters.

44%

of revenue-at-risk captured in top 20%

vs 28%

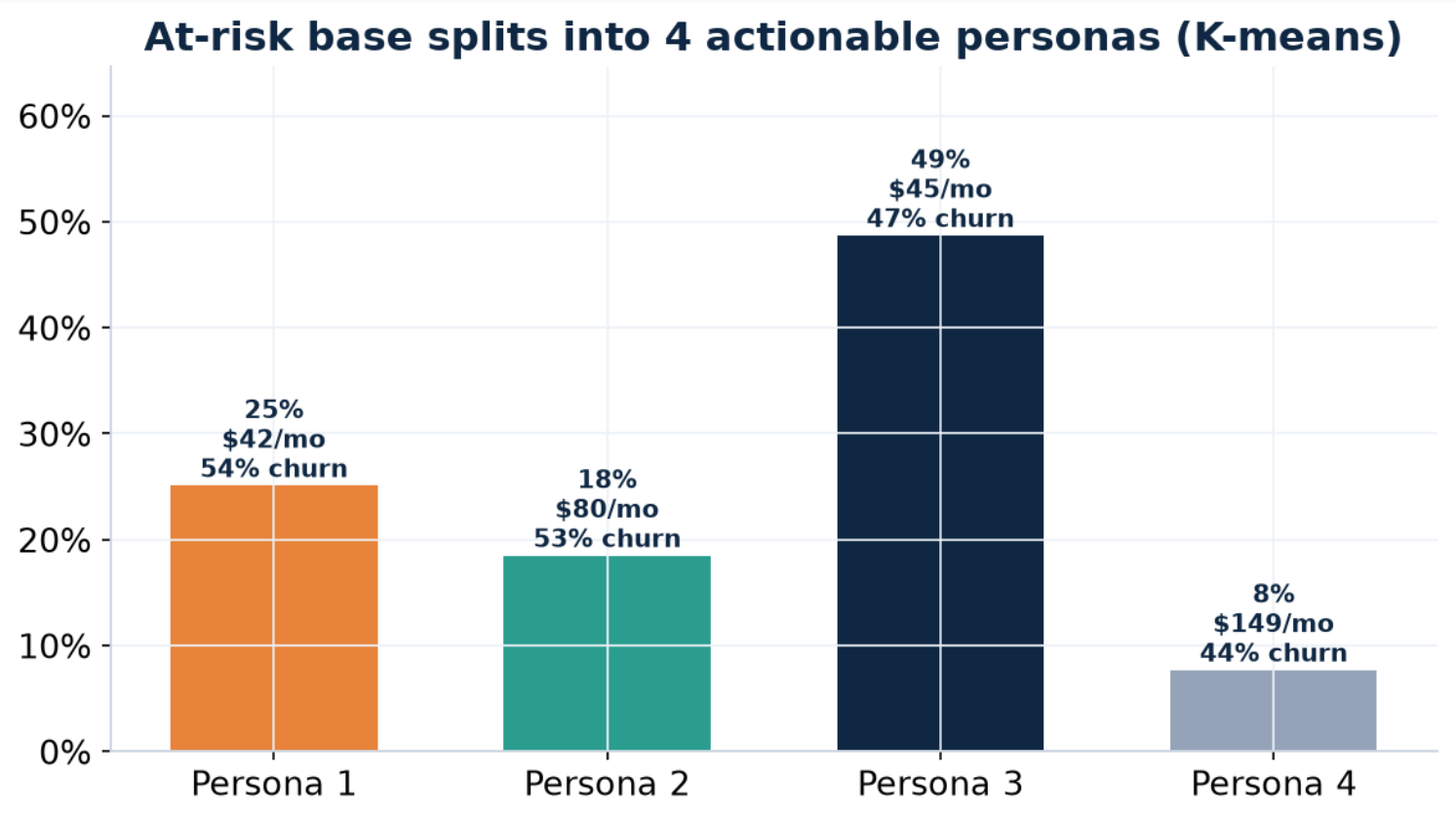
churn-only ranking

1.6x more revenue protected

Same campaign budget, same 20% of customers contacted - but we shield 1.6x the revenue by leading with value.

Source: gains simulation on held-out test set; EVaR = calibrated $P(\text{churn}) \times \text{annualised customer revenue}$.

Who to target: four data-derived retention personas



Aging-Handset

25% · \$42/mo · 54% churn

Old device, modest spend
-> Proactive upgrade / trade-in

Bill-Shock / Overage

18% · \$80/mo · 53% churn

High bill, heavy overage
-> Right-size plan + autopay discount

New / Mainstream

49% · \$45/mo · 47% churn

Newer, mid-spend, largest group
-> Onboarding nudges + loyalty perks

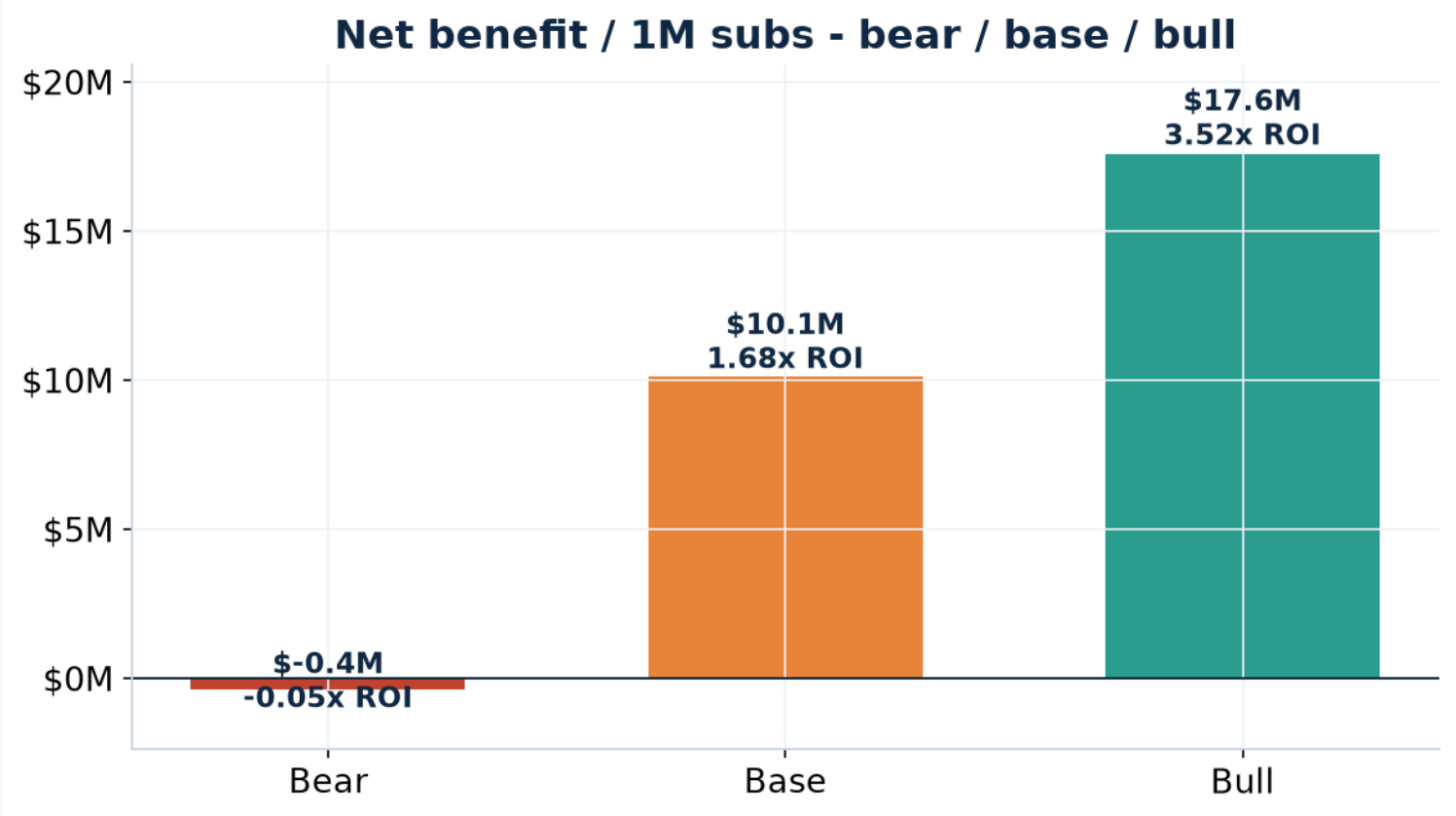
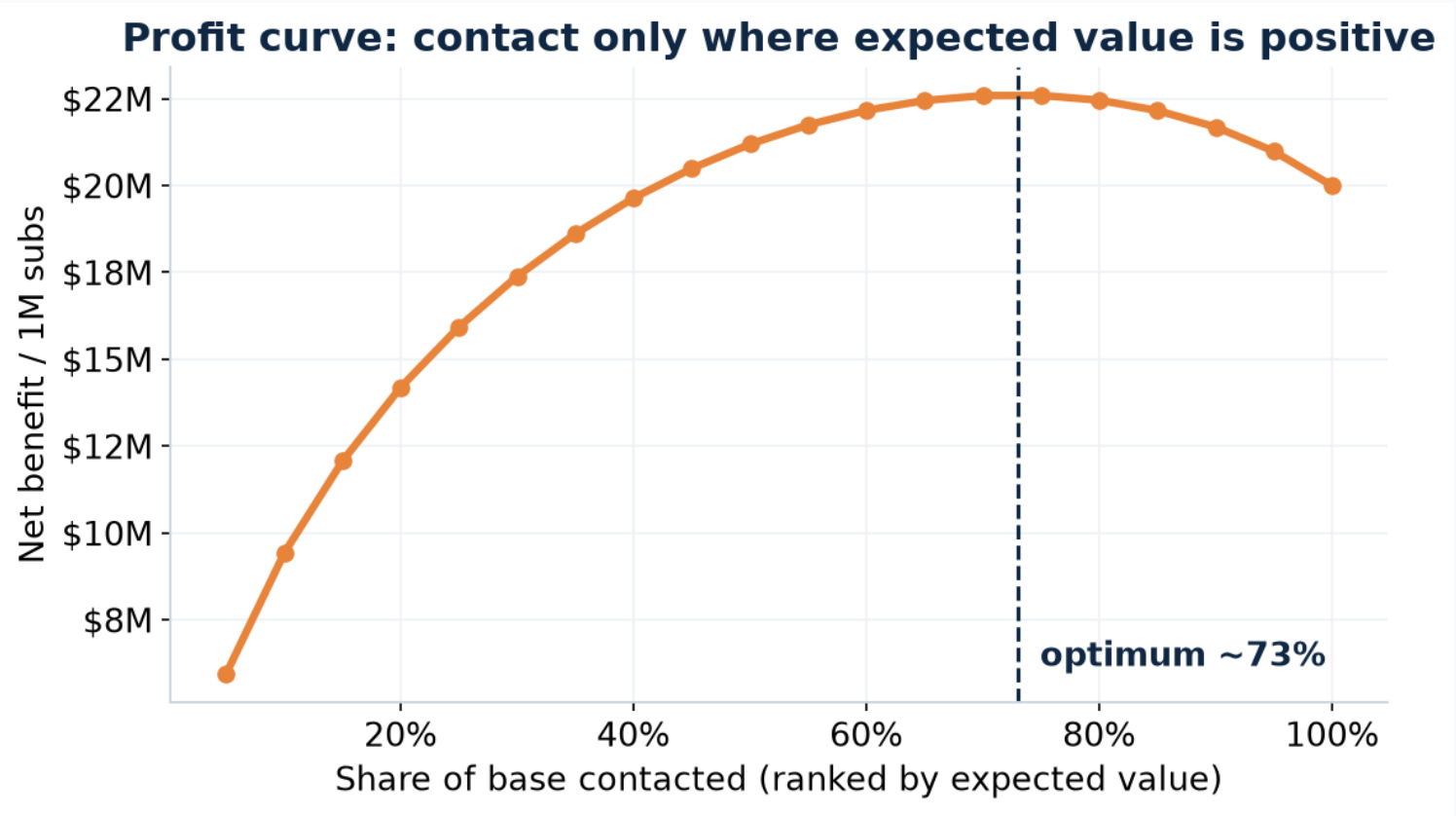
High-Value Power User

8% · \$149/mo · 44% churn

Top spend & usage
-> White-glove retention, protect margin

Source: K-means (k=4) on standardised behavioural features of the customer base, Company A dataset.

The business case: ROI-positive across scenarios



\$10.1M

Net benefit / yr

1.68x

Return on spend

16,949

Customers saved / yr

\$17.6M

Bull-case upside

Base case: 22% annual churn, 50% margin, 30% save-rate, \$30 offer, \$350 CAC. Bear case (-0.05x) flags the risk; the A/B pilot resolves which scenario is real before scaling.

Our proposal & quotation

We propose a 12-week build of the retention engine, then an ongoing managed service. Our fee is a small fraction of the value we protect.

Phase 1 · Build (wk 1-6)

Data pipeline, model, calibration, EVaR scoring, persona logic

Phase 2 · Pilot (wk 7-12)

A/B holdout on top-20% pool; validate incremental saves & ROI

Phase 3 · Scale (ongoing)

Monthly scoring, offer optimisation, drift monitoring, QBRs

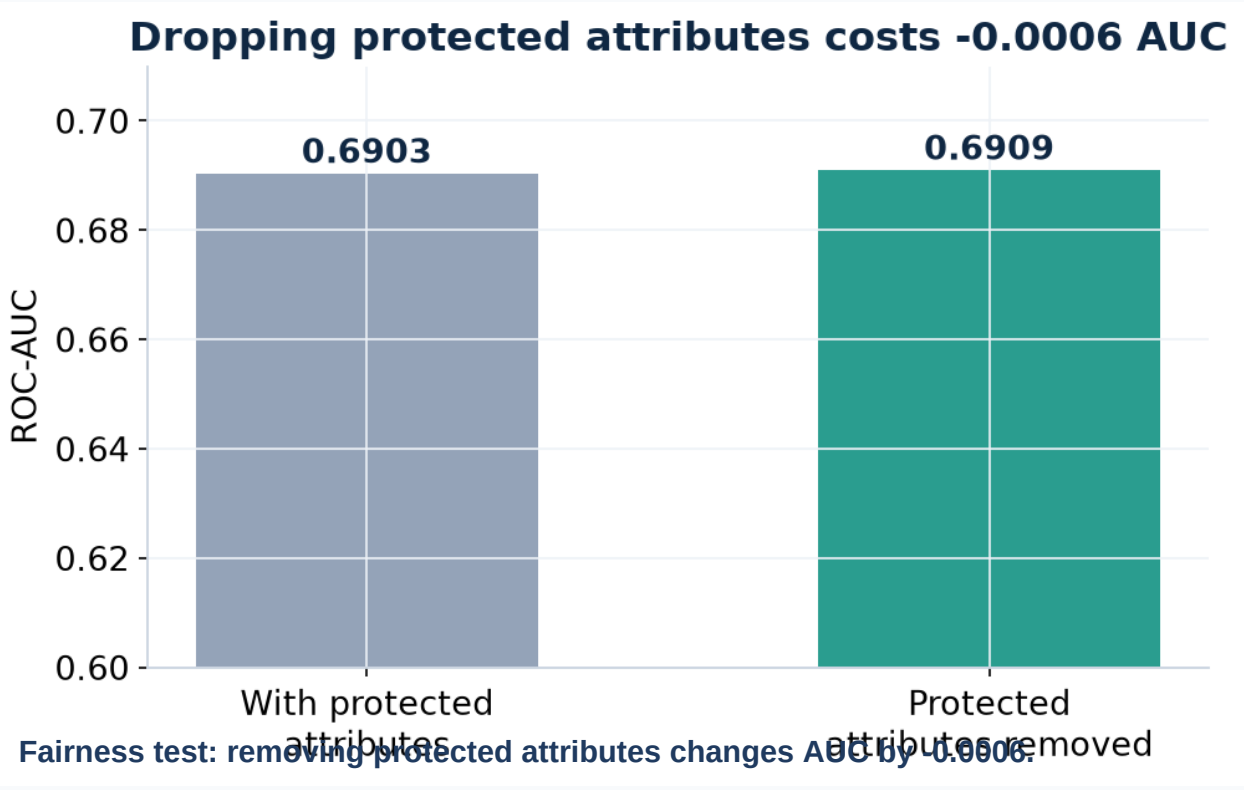
Investment

One-time setup & build	\$300k
Managed service / month	\$45k
Total year 1	\$840k

= just 8.3% of the \$10.1M net value we create

Company A keeps ~\$9.3M / yr per 1M subscribers

Fee illustrative for a PoC engagement; scales with subscriber base. Value figures per 1M subscribers / yr.



AI-use disclosure: generative AI assisted with code scaffolding, drafting and visualisation. All analysis, numbers and conclusions were computed from the dataset and verified by the author. Every figure is reproduced by the accompanying notebook.

Fairness by design

We exclude ethnicity, marital status, income & credit class. The cost is 0.06 AUC points (essentially zero) - so we lose nothing by being fair.

Validate before scaling

The 50/50 sample inflates apparent churn; the A/B pilot calibrates real save-rates and ROI - and tells us whether we are in the bear or bull scenario.

Monitor for drift

Re-train on fresh data and watch feature drift; a stale model silently loses accuracy as customer behaviour shifts.

Don't over-discount

Profit-curve guardrails and holdouts prevent giving margin away to customers who would have stayed anyway.

- Reichheld & Sasser, 'Zero Defections', HBR. · Bain & Company retention research.

- Verizon / T-Mobile investor reports 2024-25. · CTIA Wireless Survey 2025.

- McKinsey, 'AI-driven profit-driven churn', EJOR. · Lundberg & Lee (2017), SHAP, NeurIPS.

- scikit-learn; XGBoost; LightGBM documentation.

Selected references